Journal of Mathematical Sciences, Vol. 196, No. 2, January, 2014

DONAGHEY'S TRANSFORMATION: AN ELEMENTARY APPROACH

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UDC 519.12

In this paper, we study the properties of the orbits of a special transformation of plane planted trees introduced by R. Donaghey and its remarkable behavior. We prove some nontrivial properties of this transformation (the behavior of the transformation of fringed trees, "carousel effect") and obtain a lower bound on the number of its orbits. Bibliography: 7 *titles.*

1. Introduction

The transformation of plane trees considered in this paper was apparently first introduced by R. Donaghey [1]. Besides giving its definition, he mentioned the main natural problems and reduced the general problem of finding the lengths of the orbits of this transformation to considering orbits of a special class, the so-called *primitive trees*, which he related to the Motzkin numbers. As far as we know, R. Donaghey never returned to this topic. Later, this transformation was studied by A. Karttunen, who obtained a lot of empirical data presented in the encyclopaedia of integer sequences (see [5]) and stated several conjectures. In 2006, the transformation was mentioned by D. Callan [2]. However, he just acknowledged its complexity and turned to studying a simpler transformation. Any other papers on this transformation are not known to the authors.

The first author independently gave a definition of this transformation in [3], where it is called the "fractal rotation of plane trees" (the name was suggested by K. P. Kokhas). The paper contained the first and the second definitions of Donaghey's transformation (in the reverse order) and discussed the carousel effect and the alternating invariant (Theorems 4.1, 4.3 and Corollary 4.5).

Later, the first author published Theorems 6.1 and 6.2 of this paper (without the corresponding bound on the number of orbits) in [4].

This paper is a kind of "intermediate summary" and contains a systematic discussion of properties of the transformation under study (which will be called "Donaghey's transformation" or "fractal rotation of plane trees") whose proofs do not require applying advanced theories.

2. Basic definitions

Donaghey's transformation act on the set of plane planted cubic trees (abbreviated to PPCTs; a standard reference for PPCTs is, e.g., [6] or [7]) and can be naturally translated to all other combinatorial interpretations of the Catalan numbers.

Recall the basic elements of the structure of a PPCT.

(1) Every PPCT has a distinguished vertex, called its root, that has exactly one son.
(2) All the other vertices of a PPCT break into two disjoint classes: leaves and nonleaves. Every nonleaf has exactly two sons, the left and the right. A leaf has no sons and descendants.

Let α_0 be the son of the root of a PPCT T, α_1 be the right son of α_0, α_2 be the right son of α_1, etc., α_n being the right son of α_{n-1} and a leaf. The sequence $(\alpha_0, \alpha_1, \ldots, \alpha_n)$ will be called the *right major chain*. All vertices of this chain except α_n are, in turn, the roots of daughter trees (the son of the root of the daughter tree with root α_j is the left son of α_j). The daughter tree with root α_i will be denoted by $T_i = T_i(\alpha_i)$, and the whole situation will be described by the formula

$$T = |T_0(\alpha_0), T_1(\alpha_1), \ldots, T_{n-1}(\alpha_{n-1}), \alpha_n\rangle, \tag{1}$$

which will be called the *right decomposition* of the tree T.

In a similar way we define the *left decomposition* of T:

$$T = \langle\beta_0, T_1(\beta_1), \ldots, T_m(\beta_m)|, \tag{2}$$

where β_0 is a leaf and β_m is the son of the root.

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Translated from *Zapiski Nauchnykh Seminarov POMI*, Vol. 411, 2013, pp. 148–177. Original article submitted December 19, 2012.

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Definition 2.1 (inductive)**.**

(1) The base. If T is the trivial tree consisting of the root and its son, then $T' = T$.

(2) The induction step. Let T be a PPCT, and assume that for all PPCTs S with less vertices their images S' are already defined. Let $T = |T_0(\alpha_0), T_1(\alpha_1), \ldots, T_{n-1}(\alpha_{n-1}), \alpha_n\rangle$ be the right decomposition of T. Then the image of T is the tree T' with the left decomposition $T' = \langle \alpha_0, T'_0(\alpha_1), \ldots, T'_{n-1}(\alpha_n)|$.

In this form, Donaghey's transformation is defined as a transformation of a labelled PPCT (whose vertices and edges are all distinct).

Figure 1 shows the image of a labelled PPCT under Donaghey's transformation.

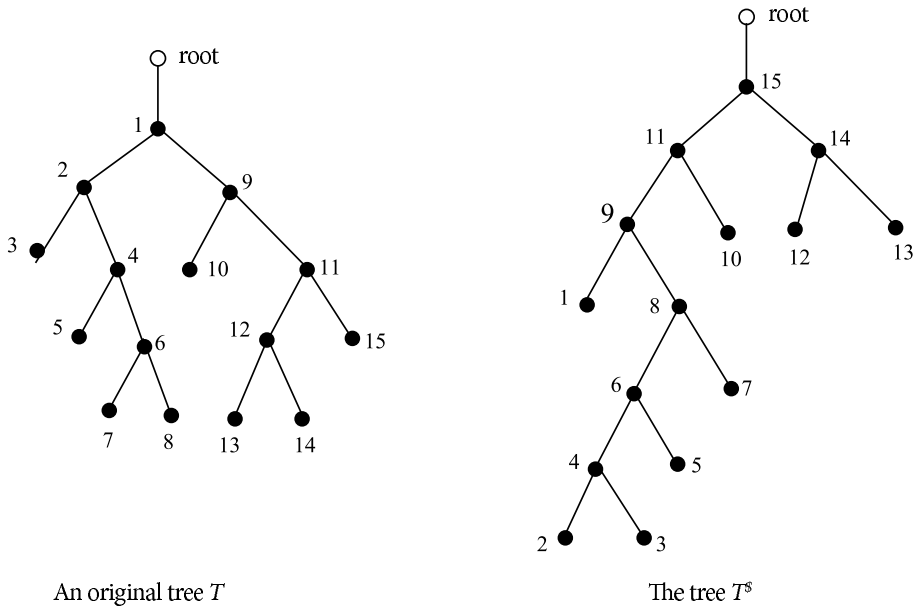


Fig. 1. Donaghey's transformation.

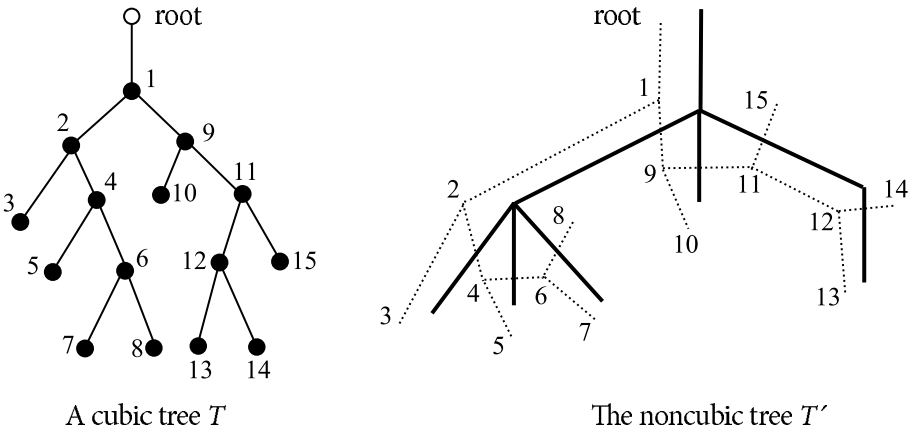


Fig. 2. The right bijection between PPCTs and PPTs.

Another definition of Donaghey's transformation appears naturally if one considers a bijection between PPCTs and plane (not necessarily cubic) planted trees (PPTs). The difference between these kinds of trees is that a nonleaf of a PPT may have any (positive integer) number of ordered sons (for definiteness, we order them from left to right; thus, when considering a PPCT as a PPT, the left son is the first and the right one is the last).

It is known that the number of PPCTs with n leaves equals the number of PPTs with n nonroot vertices; in fact, these are two of the numerous definitions of the Catalan numbers. A bijection proving this equality is also

well known. Under this bijection, the vertices of a cubic tree correspond to the *corners* (between vertices) of a noncubic tree. It is shown in Fig. 2.

Apart from the above version (which will be called *right*), this bijection has also the symmetric (*left*) version, which sends a tree T to a tree ${}^{\$}T$ (see Fig. 3).

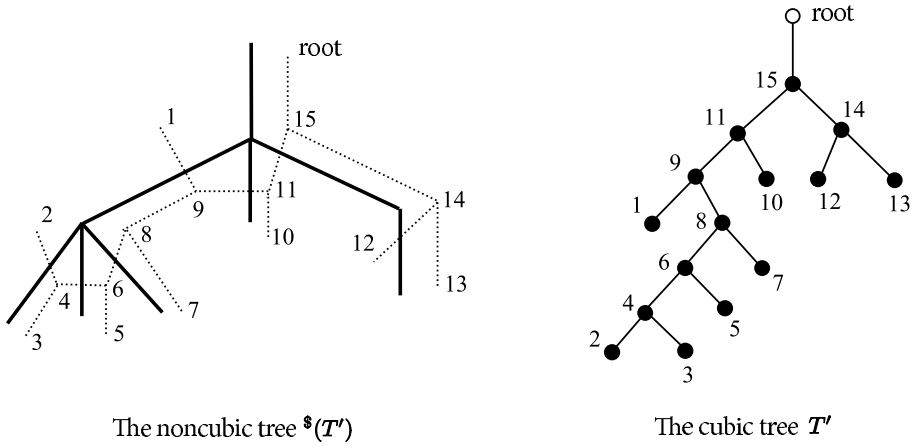


Fig. 3. The left bijection between PPCTs and PPTs.

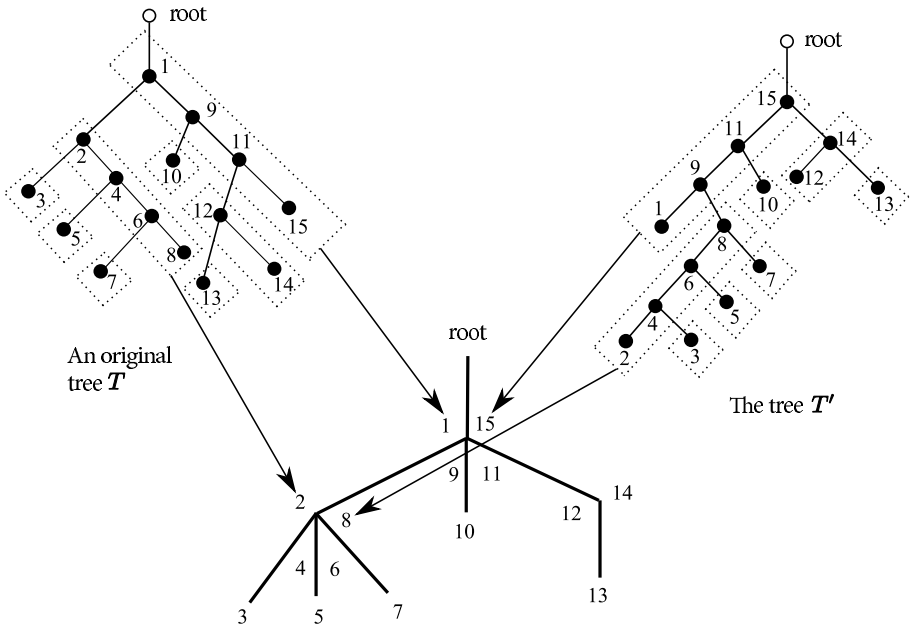


Fig. 4. The second definition of Donaghey's transformation.

The coincidence of the right trees in Figs. 1 and 3 is not fortuitous.

Definition 2.2. Given a PPCT T, let $T^{\$}$ be the PPT obtained from T by the right bijection. Let T' be the PPCT obtained from $T^{\$}$ by the inverse to the left bijection. By definition, T' is the image of T under Donaghey's transformation.

The equivalence of the two definitions of Donaghey's transformation is almost obvious. Under the inverse right bijection, the vertices of the noncubic tree turn into the right paths in the cubic tree; under the inverse left bijection, they turn into the left paths; thus their lengths coincide, and the descendant trees are in the same order and undergo similar transformations. This is illustrated in Fig. 4.

Note that the second definition allows one to trace the motion of not only vertices, but also edges of a cubic tree. In Figs. 2 and 3, one clearly sees that to each edge of the noncubic tree there correspond two edges of both cubic trees (T and T'), the "parallel" one and the "perpendicular" one. It is natural to think that the edges of T go to the corresponding edges of T'.

The third definition decomposes Donaghey's transformation into the product of two involutions and relates it to another well-known interpretation of the Catalan numbers, the so-called *Dyck paths*.

Namely, on the set of all PPCTs there is a very natural involution σ (which will be called the transposition), which sends all left sons to right ones and vice versa. On the set of Dyck paths there is also a natural involution θ (which will be called the reflection), which corresponds to going backwards along a path (Fig. 5). Besides, there is a canonical bijection φ that sends a PPCT with n leaves to a positive path with $2(n-1)$ steps.

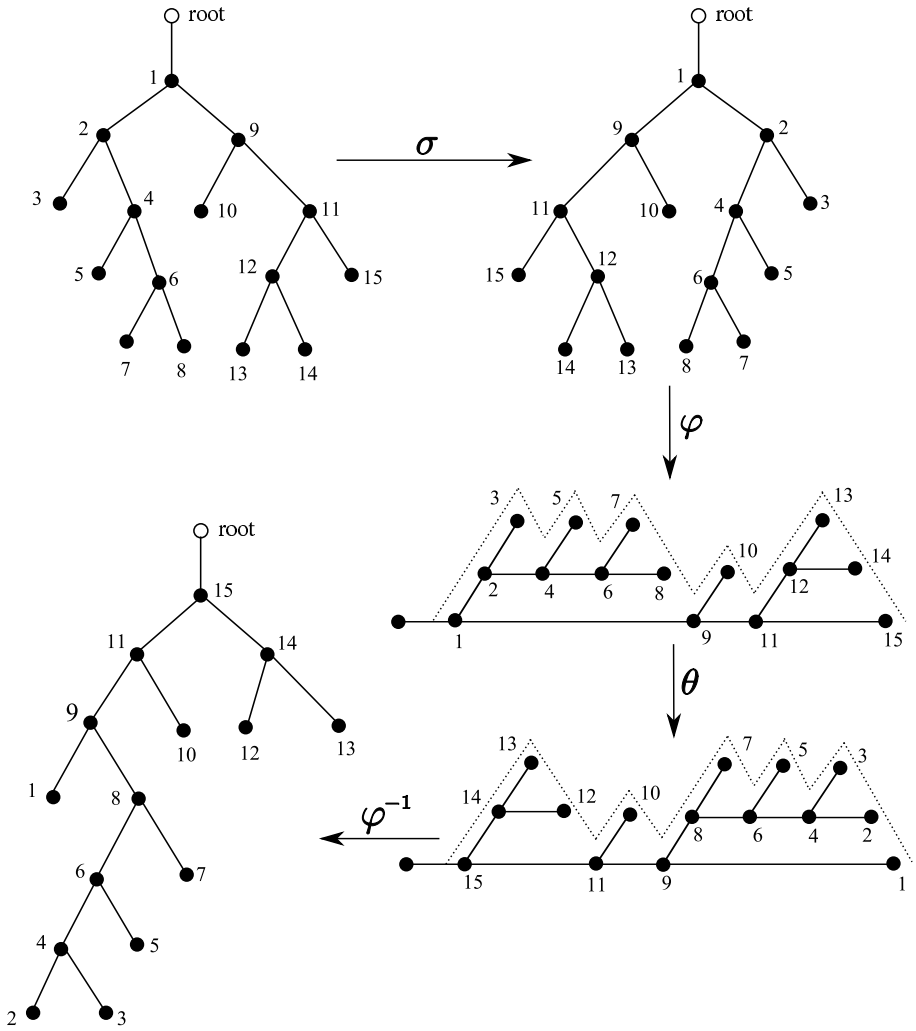


Fig. 5. The third definition of Donaghey's transformation.

Definition 2.3. Donaghey's transformation is the composition $\varphi^{-1} \circ \theta \circ \varphi \circ \sigma$ (the transformations are applied from right to left).

The equivalence of the third definition to the second one follows from the explicit form of the bijection between positive paths and noncubic trees illustrated in Fig. 6.

Thus we have proved the following theorem.

Theorem 2.1. *The above three definitions are equivalent.*

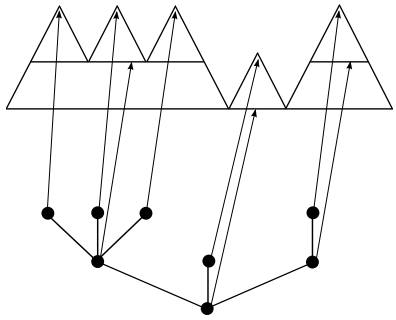


Fig. 6. The bijection between positive paths and PPTs.

3. Donaghey's transformation in terms of other combinatorial interpretations of the Catalan numbers

In terms of plane trees, Donaghey's transformation looks rather complicated. However, one could imagine another combinatorial interpretation of the Catalan numbers for which its properties would be more transparent.

Strictly speaking, we *cannot* claim with certainty that such an interpretation does not exist. Formally, one could consider, for instance, the set of orbits of Donaghey's transformation on the set of trees of fixed size, regarding them as polygons with labelled vertices (where a vertex is labelled by the tree it corresponds to); every polygon has a distinguished vertex (corresponding to the current tree). However, although in this interpretation the properties of Donaghey's transformation are transparent, it is itself extremely unnatural and obscure. For example, it is not clear how many and which polygons it contains (for a given size of trees).

Let us describe Donaghey's transformation in terms of two most important combinatorial interpretations of the Catalan numbers (besides trees): Dyck paths and triangulations of polygons (with a distinguished side corresponding to the root of a PPCT).

According to the third definition, to describe Donaghey's transformation in terms of Dyck paths, it suffices to describe the transposition σ of PPCTs (see Figs. 7 and 8).

We will give an inductive description of this transposition in terms of Dyck paths. The base (for paths of length 2) is obvious.

Further, the zero level vertices split every Dyck path into *irreducible subpaths*.

If there are several such subpaths, choose the first of them. (For the time being, we exclude from consideration the case where such a subpath is unique.) Removing from it the first and the last steps, we again obtain a Dyck path, which we denote by P. The case where P is trivial will also be considered later. Denote the remaining part of the Dyck path under consideration by Q. The transformation of paths corresponding to the transposition of PPCTs consists in applying the transformation to the (simpler) paths P and Q *separately*, obtaining paths P' and Q', and then swapping them (see Fig. 7).

In the case where an irreducible subpath is unique (this case will be called *special*), the path Q is trivial. Such a path corresponds to a cubic tree for which the subtree of left descendants of the root is trivial. A tree for which the subtree of right descendants of the root is trivial corresponds to a Dyck path that begins with a peak of height 1, for which the subpath P is trivial. Therefore, in the special case, the analog of the transposition of trees looks as in Fig. 8, which describes the two cases we have not yet considered.

Applying the reflection θ to the paths obtained above, we get an inductive description of Donaghey's transformation in terms of Dyck paths. Note that Donaghey also considered the transformation in terms of *bracketings*, which are very close to Dyck paths. However, this was just an auxiliary observation (when passing to primitive trees) and did not result in a description of the transformation in these terms.

In terms of triangulations of polygons, the counterparts of right (respectively, left) sequences of descendants are *caterpillars* – triangulations in which all triangles share a common vertex to the right (respectively, left) of the distinguished side of the polygon. Correspondingly, one should split a triangulation into caterpillars and then move the distinguished side in each caterpillar one position clockwise. Figure 9 shows an example of the decomposition into caterpillars and the subsequent rotation. For clarity, instead of a noncubic tree we have taken a "cactus," which is another combinatorial interpretation of the Catalan numbers (almost identical to noncubic trees). The caterpillars of the triangulation correspond exactly to the lower sides of the triangles of the cactus.

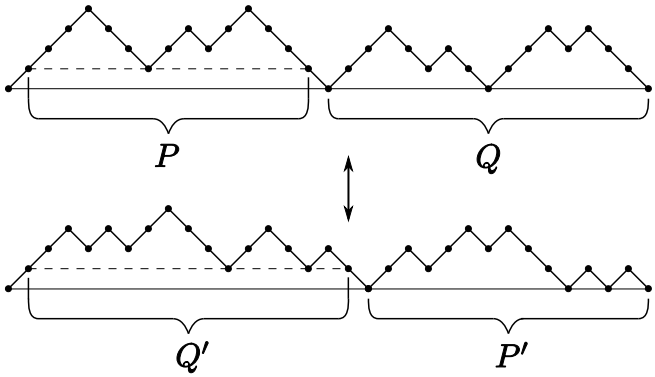


Fig. 7. The transposition of PPCTs in terms of Dyck paths: the regular case.

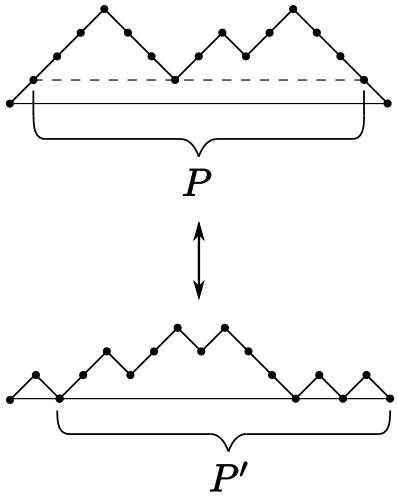


Fig. 8. The transposition of PPCTs in terms of Dyck paths: the special case.

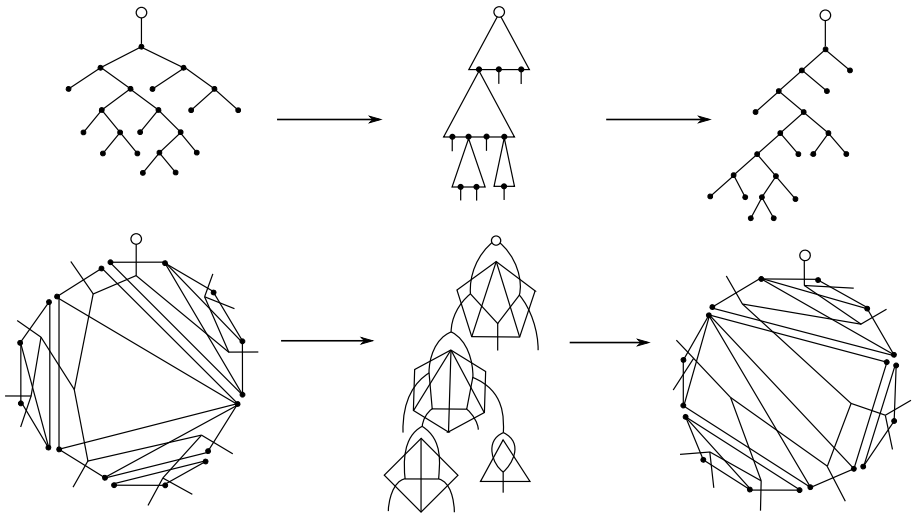


Fig. 9. Translating Donaghey's transformation to triangulations of polygons.

Putting aside trees and cacti, we arrive at the following description of Donaghey's transformation in terms of triangulations with distinguished side (see Fig. 10).

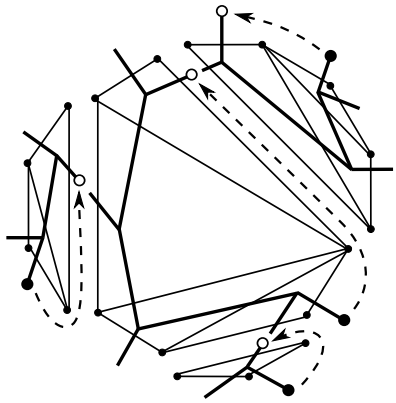


Fig. 10. Donaghey's transformation in terms of triangulations of polygons.

(1) Take the caterpillar corresponding to the right vertex of the distinguished side of the triangulation.
(2) Remove it from the triangulation; the remaining part of the triangulation is a disjoint union of several smaller triangulations; as the distinguished side of such a triangulation we take the side adjacent to the removed caterpillar.
(3) Apply the previous two steps to the obtained triangulations; repeat this procedure until the triangulation splits into caterpillars.
(4) In each of the obtained caterpillars, move the distinguished side one position clockwise.
(5) Assemble the obtained triangulation.

We cannot mention all known combinatorial interpretations of the Catalan numbers. However, this is not necessary, since, although numerous, they are usually very close either to one of the two types of trees we have considered (for example, binary trees, cacti, nonassociative products), or to Dyck paths (bracketings, arc diagrams, various paths, parallelogram polynominoes), or to triangulations of polygons (positive definite three-diagonal matrices of positive integers with determinant 1 and ones outside the main diagonal). It remains to say that a fine collection of combinatorial interpretations of the Catalan numbers is given in [7].

The main conclusion of this section is that in all principal combinatorial interpretations of the Catalan numbers, Donaghey's transformation exhibits roughly the same behavior: it is more or less natural, but distinctly "fractal," so that its properties are not clear.

4. The carousel effect

We can consider a more detailed classification of nonroot vertices of a PPCT by dividing them into five types: the one-element class consisting of the son of the root (in what follows, SR); the left leaves (LL), i.e., leaves that are left sons; the right leaves (RL); the left and right nonleaves (LN and RN, respectively). For example, in the tree T shown in Fig. 1, vertex 1 is SR, vertices 3, 5, 7, 10, 13 are LL, vertices 8, 14, 15 are RL, vertices 2 and 12 are LN, vertices 9, 4, 6, 11 are RN. Analogously, the corners of a PPT can be divided into four classes — left, right, lower, and flat — according to their form. For example, in the tree $T^{\$}$ shown in Fig. 2, corners 1, 2, 12 are left, corners 8, 14, 15 are right, corners 9, 4, 6, 11 are lower, and corners 3, 5, 7, 10, 13 are flat.

One clearly sees that when we pass from T to $T^{\$}$, SR and LN vertices turn into left corners, RN vertices turn into lower corners, LL vertices turn into flat corners, and RL vertices turn into right corners. It is equally clear that under the left inverse bijection between PPTs and PPCTs (see Fig. 3), right corners turn into SR or RN vertices (the SR vertex comes from the right corner adjacent to the son of the root of the PPT), lower corners turn into LN vertices, left corners turn into LL vertices, and flat corners turn into RL vertices. We obtain the following result.

Theorem 4.1 (The carousel effect)**.** *Let T be a PPCT and T' be its image under Donaghey's transformation.*

(1) *If α is an SR or LN vertex of T, then it is an LL vertex of T'.*
(2) *If α is an RN vertex of T, then it is an LN vertex of T'.*
(3) *If α is an LL vertex of T, then it is an RL vertex of T'.*
(4) *If α is an RL vertex of T, then it is an SR or RN vertex of T'.*

Remark. Theorem 4.1 does not say which of the possibly numerous LL vertices of T' is the image of a given LN vertex α of T. Below we refine it.

On the set of nonroot vertices of a PPCT T there are several natural traversals. The most natural ones are the left and right standard traversals. The left standard traversal can be described recursively: we start from the son of the root (or, equivalently, from the root itself); arriving at a vertex α, we first visit this vertex, then the tree of its left descendants, and, finally, the tree of its right descendants. Actually, the vertices of the tree T in Fig. 1 are numbered according to the left standard traversal. The right standard traversal is symmetric to the left one.

We may also consider the right and left inverse traversals, i.e., the same traversals passed backwards.

On the other hand, on a PPT (it is convenient to regard it as a *one-hedron*) $T^{\$}$ there are also two standard traversals: counterclockwise and clockwise along the "perimeter." One may assume that all traversals start at the root. We say that a traversal of the vertices of a PPCT T is the *left* (respectively, *right*) *natural* traversal if it agrees with the counterclockwise (respectively, clockwise) traversal of the PPT $T^{\$}$.

Theorem 4.2 (Agreement of traversals)**.** *The left natural traversal is the standard left traversal; the right natural traversal is the right inverse traversal.*

This result immediately follows from the recursive definitions of the traversals and their visualizations.

On the vertex set of a tree T, we fix the order corresponding to the standard left traversal; on the same set regarded as the vertex set of the tree T', we fix the order corresponding to the right inverse traversal. Then we number the vertices of each type separately according to these orders.

Theorem 4.3 (The refined carousel effect)**.** *Let α be a vertex of T and T'. Then*

(1) *If α is the SR vertex in T, then it is the first LL vertex in T'.*
(2) *If α is the kth LL vertex in T, then it is the kth RL vertex in T'.*
(3) *If α is the kth RL vertex in T (not the last one), then it is the kth RN vertex in T'.*
(4) *If α is the last RL vertex in T, then it is the SR vertex in T'.*
(5) *If α is the kth RN vertex in T, then it is the kth LN vertex in T'.*
(6) *If α is the kth LN vertex in T, then it is the $(k+1)$th LL vertex in T'.*

Remark. The situation is not as simple as one might think. Recall that the orders in question are *substantially different*. Hence it is not that a vertex just moves successively through all possible positions in the tree. In fact, the traversals under consideration coincide on the set of leaves, because leaves are always incomparable, i.e., no leaf is a descendant of another one. On the other hand, right (respectively, left) nonleaves form right (respectively, left) chains the order of vertices in which, as one can easily see, is inverted under each transformation. Besides, chains of nonleaves *are not preserved* under the iterations of the basic transformation, so that one apparently cannot predict which right nonleaf will be the image of a given vertex after several steps.

Given a tree T, let $k(T)$, $l(T)$, $m(T)$, and $n(T)$ be the number of left leaves, right leaves, right nonleaves, and left nonleaves of T, respectively.

Corollary 4.4. $k(T') = n(T) + 1$, $l(T') = k(T)$, $m(T') = l(T) - 1$, $n(T') = m(T)$.

Note that the following equalities obviously hold in every PPCT T: $k(T) = m(T) + 1$, $l(T) = n(T) + 1$.

Correspondingly, in order to determine the quadruple $(k(T), l(T), m(T), n(T))$, it suffices to specify the *type of the tree*, i.e., the pair $(k(T), l(T))$.

Definition 4.1. We say that PPCTs T_1 and T_2 are equivalent if each of them can be obtained from the other one by several iterations of the basic transformation; in this case, we write $T_1 \approx T_2$.

Corollary 4.5 (The alternating invariant)**.** *If $T_1 \approx T_2$, then $(k(T_1), l(T_1)) = (k(T_2), l(T_2))$ or $(k(T_1), l(T_1)) = (l(T_2), k(T_2))$.*

Remark. Simple examples show that this invariant is not complete, i.e., there exist nonequivalent PPCTs with equal subtypes.

Denote by $T(k, m)$ the set of all PPCTs with k left leaves and m right leaves. According to Corollary 4.5, the orbit of every representative of $T(k, m)$ under Donaghey's transformation is contained in $T(k, m) \cup T(m, k)$. Obviously, all these classes are nonempty, hence every such union contains at least one orbit. Therefore, for odd

n, the set $T(n)$ of trees with n leaves breaks into at least $\frac{n-1}{2}$ orbits; for even n, it breaks into at least $\frac{n}{2}$ orbits, since the symmetric class $T(\frac{n}{2}, \frac{n}{2})$ also contains at least one orbit. On the other hand, for odd n the lengths of all orbits are even, and for even n the orbits of odd length can be contained only in the symmetric class.

5. Donaghey's transformation is hyperbolic

The hyperbolicity of a transformation means that a small change in the initial point drastically changes the result after several iterations of the transformation. In our case this means that the transformation is not monotone with respect to the standard lattice ordering of all trees (recall that the PPCTs form a distributive lattice whose irreducible elements are ordered as the vertices of the infinite cubic tree). This fact itself is rather obvious, but it is of great importance, being the main reason for considering Donaghey's transformation.

Definition 5.1.

(1) Let α be a right edge of a PPCT T, joining a father a with its right son b. Consider the new PPCT $T(\alpha)$ that differs from the original one in that
   (a) $T(\alpha)$ has two vertices (say, c and d) and two edges (say, β and γ) more;
   (b) the edge α joins the father a with the right son c;
   (c) the edge β joins the father c with the right son b;
   (d) the edge γ joins the father c with the left son d, which is a leaf.

(2) Passing from T to $T(\alpha)$ will be called partitioning the edge α (Fig. 11).

(3) The operation of partitioning a left edge is defined in a completely similar way.

(4) A possibility to partition an edge is a function of this edge, which we will call the *germ* corresponding to this edge.

(5) It may happen that for an edge α of a tree T there exists an edge β of the tree T' such that $(T(\alpha))' = T'(\beta)$. In this case, we say that Donaghey's transformation moves the germ from the edge α to the edge β. In particular, if $\alpha = \beta$, we say that the germ stays on the edge α.

(6) If (5) is not the case, we say that the germ *collapses*. This is the simplest manifestation of the hyperbolicity of Donaghey's transformation.

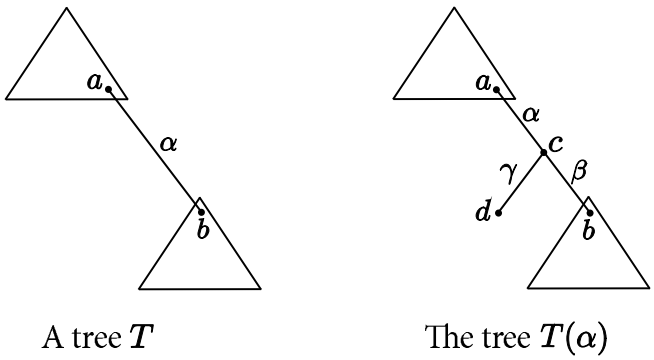


Fig. 11. Partitioning an edge.

Edges of a PPCT not adjacent to its root (we will call them nonroot edges) are also naturally divided into leaves and nonleaves. In what follows, given an edge α that is a right nonleaf joining a vertex a with its right son b, also a nonleaf, we define the next edge as the edge joining b with its right son c.

Theorem 5.1 (Germ moving rules)**.** *Under Donaghey's transformation,*

(1) *a germ incident to a left edge stays on it (and it becomes a right edge);*

(2) *a germ from a right nonleaf moves to the next edge;*

(3) *a germ from a right leaf collapses.*

The theorem follows almost immediately from the first definition of Donaghey's transformation. Claim 3 is illustrated in Fig. 12. However, the situation shown in the right part of Fig. 12 could be interpreted as "a right germ on a right edge." For a more convincing illustration, let us repeat the transformation.

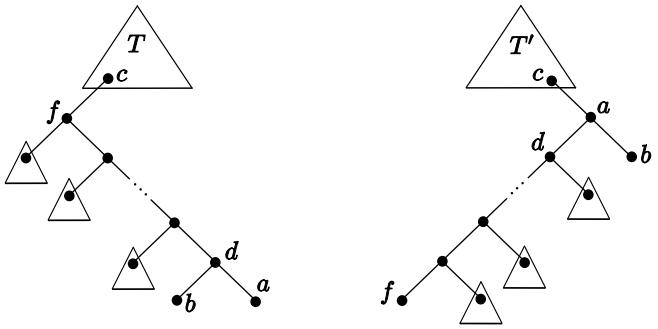
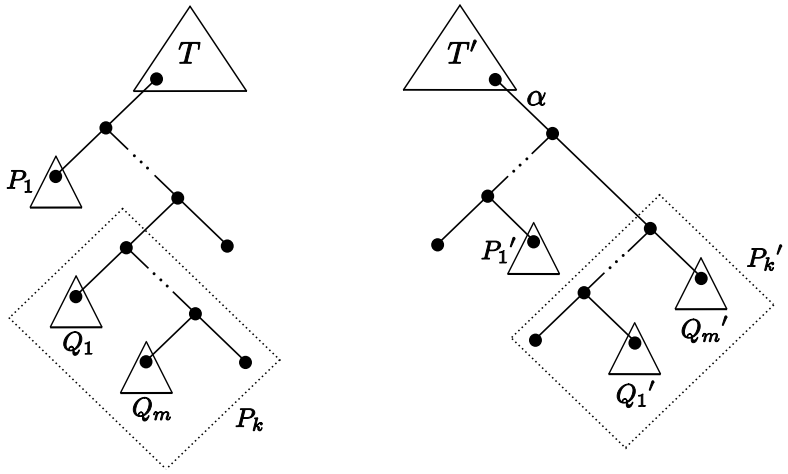
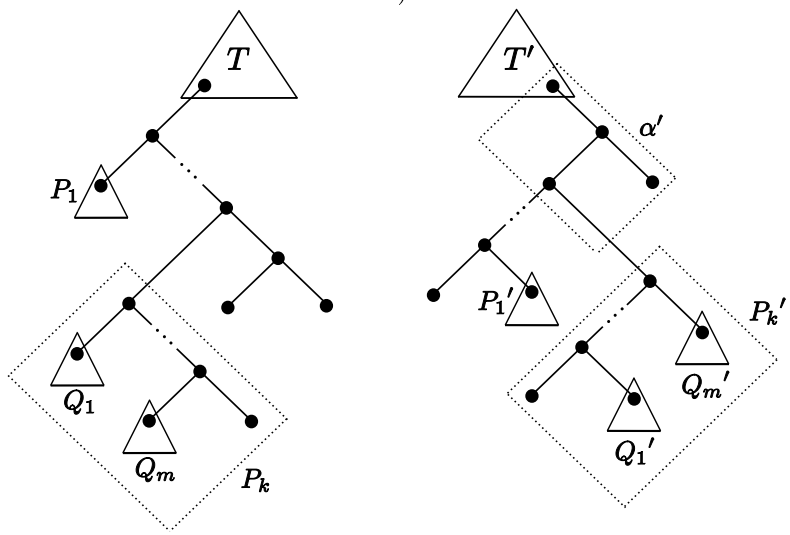


Fig. 12. A germ on a right leaf collapses.



a)



b)

Fig. 13. The first iteration of Donaghey's transformation a) of T, b) of $T(\alpha)$.

According to the same first definition, the behavior of left subtrees under Donaghey's transformation is *autonomous*, i.e., independent of the other vertices and edges in the tree. Hence it suffices to trace only the behavior of the "nearest neighborhood" of the germ under consideration. Comparing the right parts of Figs. 13 a) and 13 b), we see that the only difference between the first Donaghey's transformation of T and that of $T(\alpha)$ is

that the edge α is replaced by the fragment α'. Hence we may assume that before making the second iteration we have the picture shown in the left parts of Figs. 14 a) and 14 b), respectively. But comparing the right parts of Figs. 14 a) and 14 b), we see that the structure of the tree has *drastically* changed.

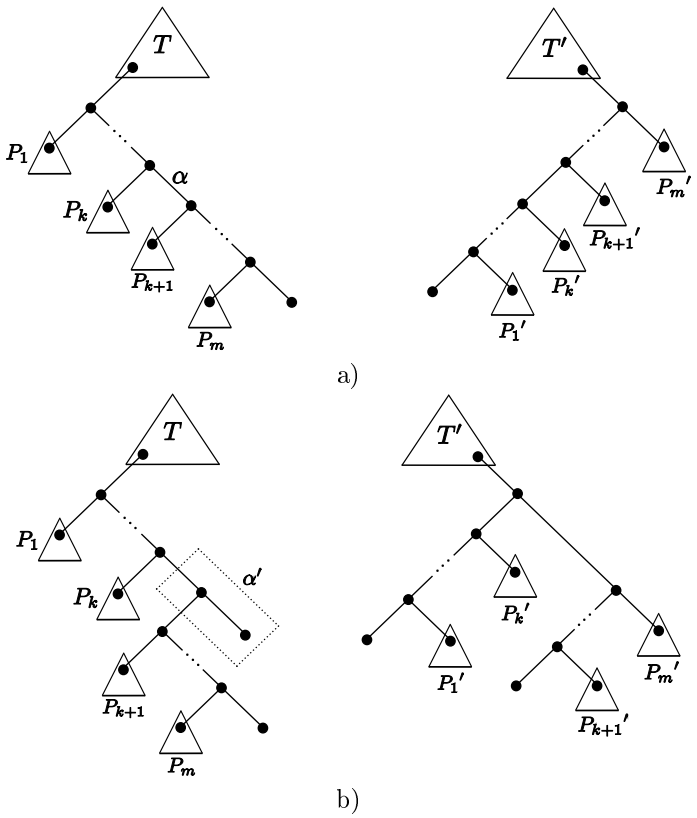


Fig. 14. The second iteration of Donaghey's transformation a) of T, b) of $T(\alpha)$.

Remark. It might appear that the authors have not tried hard enough: the right germ on the right edge has collapsed and turned into some other element of the structure of the tree whose evolution can also be traced until it turns back into a germ; and throughout this evolution, the germ does not turn into something too complicated.

The authors got the impression (difficult to formulate as a precise conjecture) that this is *absolutely not the case*. Conversely, if the size of the tree and the number of iterations grow, a germ may turn into arbitrarily complicated elements. It is this intriguing fact that makes it impossible to construct a "tame" theory of Donaghey's transformation.

6. The fringing theorem

Consider another operation on PPCTs, which will be called *fringing*.

Definition 6.1.

(1) Let T be a PPCT and a be a (*vertex*) leaf of T. Add two new vertices a_1 and a_2, making them the left and the right son of a, respectively. Performing this transformation for each leaf of T, we obtain a new tree $Q(T)$, which will be called the fringing of T.

(2) If a tree T is the fringing of another tree, it will be called fringed (Fig. 15).

By $\Delta(T)$ we denote the length of the orbit of a tree T. The following theorem holds.

Theorem 6.1.

(1) *If* $\Delta(T)$ *is odd, then* $\Delta(Q(T)) = 3\Delta(T)$.

(2) *If* $\Delta(T)$ *is even, then* $\Delta(Q(T)) = \frac{3}{2}\Delta(T)$.

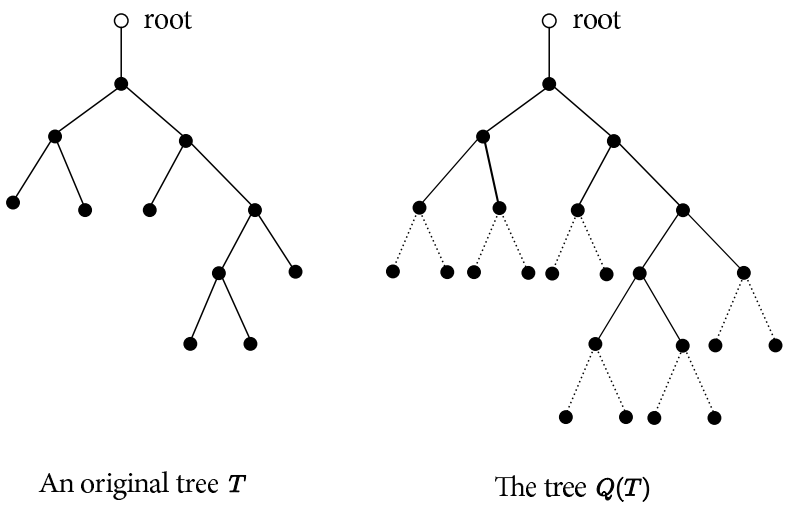


Fig. 15. The fringing of a tree.

Proof. It suffices to prove that

$$Q(T'') = (Q(T))'''. \tag{3}$$

It also follows that if $\Delta(T)$ is even, then the length of the orbit is, in a sense, tripled under fringing, but the orbit itself breaks into two parts.

To prove this equality, we need several definitions.

Definition 6.2.

(1) Let T be a PPCT. By T° we denote the PPT obtained from T by inserting a new vertex between the root and the son of the root; thus the new vertex becomes the new son of the root, and the old son of the root becomes its unique son (see Fig. 16). The tree T° is no longer cubic.

(2) A left (right) *isocubic* tree ${}^*T$ (respectively, T^*) is a PPCT that under the right (respectively, left) bijection between PPCTs and PPTs goes to a PPT of the form T° for some PPCT T.

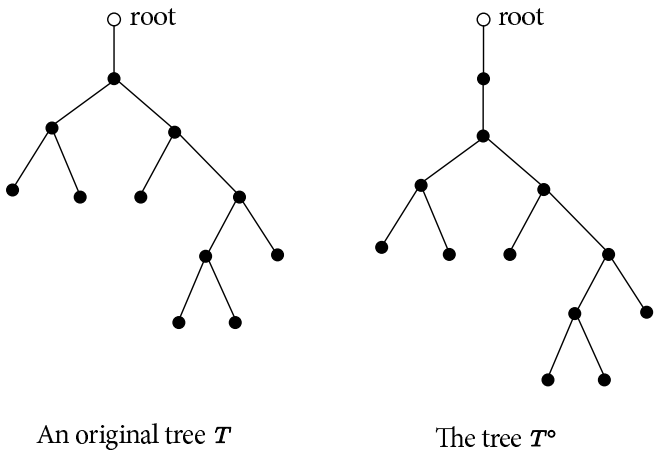


Fig. 16. Transforming T into T°.

An example of a left isocubic tree is given in Fig. 17. A characteristic feature of left (respectively, right) isocubic trees is the absence of nontrivial (of length greater than 1) chains of right (respectively, left) nonleaves (here by nonleaves we mean edges).

By the second definition of Donaghey's transformation, $({}^*T)' = T^*$.

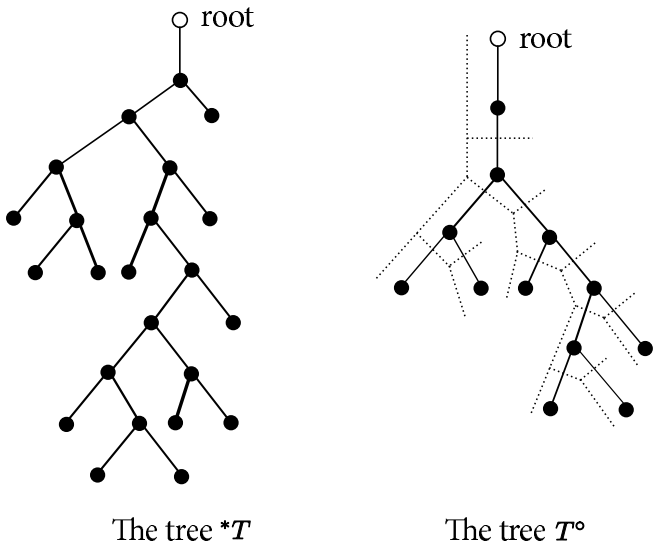


Fig. 17. An example of a left isocubic tree.

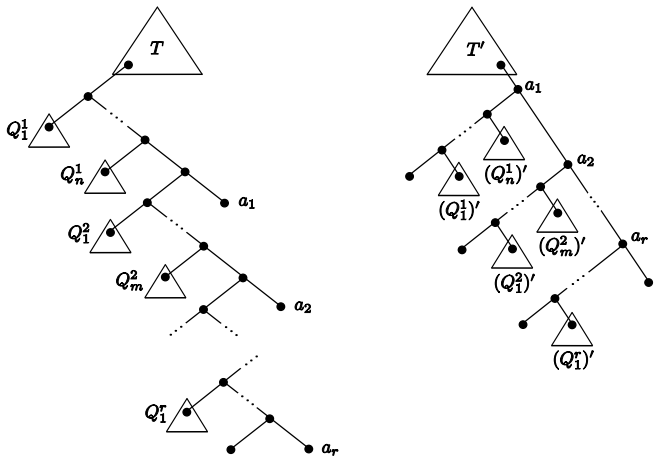


Fig. 18. Donaghey's transformation of the original tree.

To prove (3), it suffices to show that

$$(Q(T))' =^* (T'). \tag{4}$$

Indeed, by the symmetry, the fringing operation turns the chain of two successive rotations $T \to T' \to T''$ into the chain of three successive rotations $Q(T) \to^* (T') \to (T')^* \to Q(T'')$.

To prove (4), compare the transformations of the original tree and the fringed tree (see Figs. 18, 19, and 20). Figure 18 shows the transformation of the original tree. After the rotation, the vertices $a_1, a_2, \ldots, a_r$ form a right chain. Figure 19 shows the transformation of the fringed tree. The fringing breaks the right chain of vertices; one clearly sees that a left isocubic tree appears. Figure 20 shows how passing from PPCTs to PPTs recovers the right chain, which now consists of the vertices $D_1, \ldots, D_r$. □

It is interesting that the third iteration of Donaghey's transformation reveals another carousel effect. The edges of the fringed tree can be naturally divided into three classes: internal edges, regular fringes, and two singular fringes incident to the new sons of the last (with respect to the left traversal) right leaf of the original (nonfringed) tree. The corresponding partition is shown in Fig. 21 a).

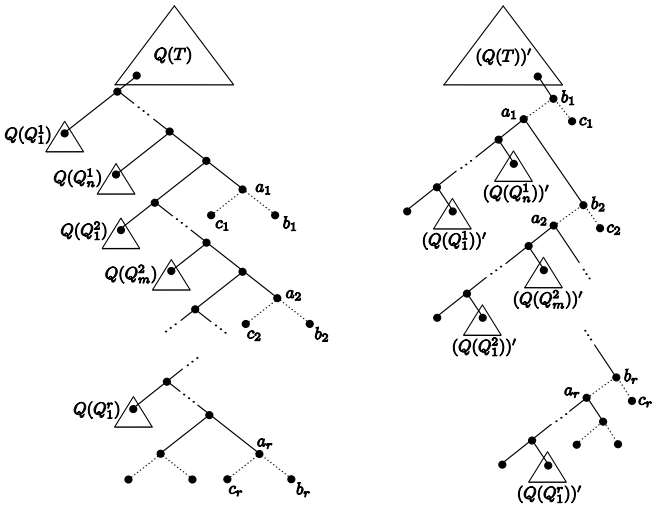


Fig. 19. Donaghey's transformation of the fringed tree.

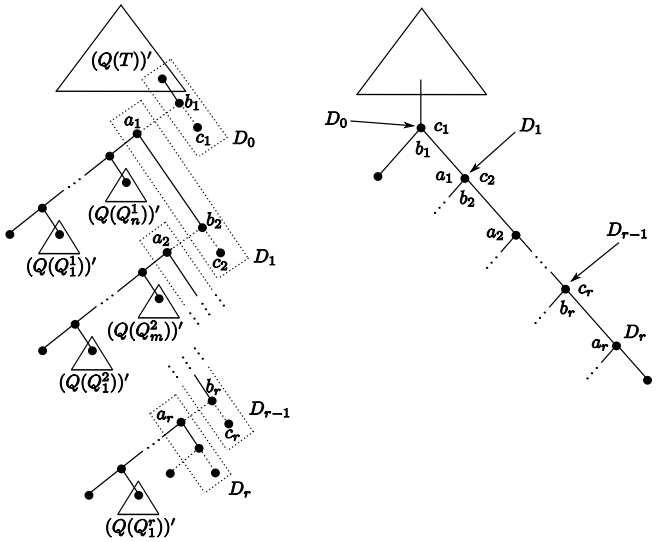


Fig. 20. Recovering the right chain $D_1, \ldots, D_r$.

Theorem 6.2 (The carousel effect for the edges of fringed trees)**.** *Under the composition of three Donaghey's transformations, internal edges become fringes, regular fringes become internal edges, and singular fringes become fringes incident to the new sons of the first (with respect to the left traversal) left leaf of the original tree.*

Proof. After the first rotation, the singular fringes move "under the root," the other fringes are located in the right and lower part in right chains, and the internal edges are located in the left and upper part. After the second rotation, the fringes and internal edges swap places. The rest follows from the symmetry. An example is shown in Fig. 21. □

It follows from Theorem 6.1 that the length of the orbit of the complete n-level tree, i.e., the tree containing all possible descendants of the son of the root up to the nth level, is equal to 3^{n-1} (we assume that the son of the root is at the zero level). Thus we have obtained the following result.

Corollary 6.3. *There exists an infinite sequence of trees for which the length of the orbit grows as* $O(n^{\log_2 3})$ *with respect to the number of leaves.*

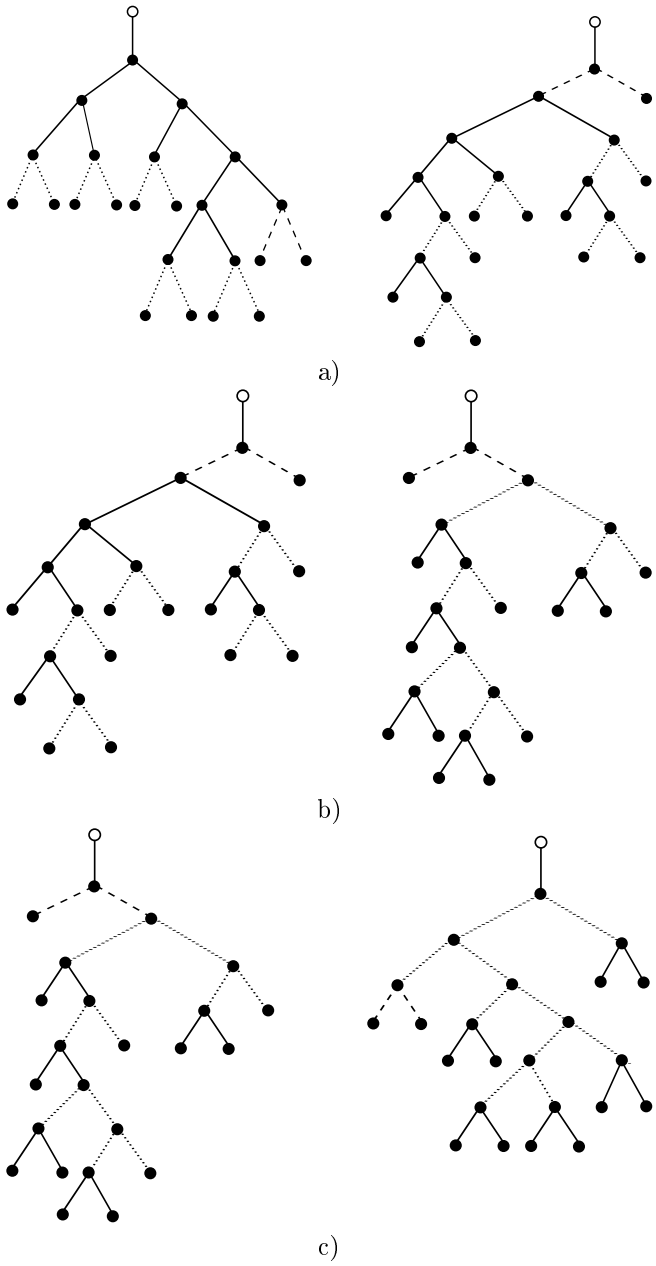


Fig. 21. The carousel effect for the edges of a fringed tree.

Remark. In fact, this is rather a mass phenomenon: instead of the complete n-level trees, one may consider the successive fringings of an arbitrary fixed tree.

The empirical data presented at the end of the paper show that this corollary can be substantially strengthened. A natural conjecture is that there exist sequences of trees for which the lengths of the orbits grow exponentially.

Besides, Corollary 6.3 shows that there exist trees with arbitrarily long orbits. Therefore, on the set of all trees Donaghey's transformation acts as the free cyclic group.

Recall that Donaghey's transformation τ is the composition of two involutions: the transposition σ of trees and the reflection θ of paths: $\tau = \theta \circ \sigma$. This allows us to state a stronger assertion.

Theorem 6.4. *The group generated by the involutions from Definition* 2.3 *is the free product* $Z_2 * Z_2 = \langle\theta\rangle * \langle\sigma\rangle$.

Proof. Since σ and θ are involutions, every relation in the group under consideration might have one of the four types: $\tau^k = e$, $\tau^{-k} = e$, $\sigma \circ \tau^k = e$, or $\tau^k \circ \theta = e$. The first two of them cannot hold for all trees, since, as mentioned above, there exist trees with arbitrarily long orbits.

Assume that $\sigma \circ \tau^k(T) = T$ for every PPCT T. Then $\tau^k(T) = \sigma(T)$, $\tau^{2k}(T) = \sigma^2(T) = T$, which contradicts the existence of arbitrarily long orbits. A similar argument applies to the last relation. □

Besides, the fringing theorem allows us to refine the estimate on the number of orbits obtained at the end of Sec. 4.

Consider the family of sequences $(a_k^{(m)})_{k=1}^{\infty}$, $(b_k^{(m)})_{k=1}^{\infty}$, and $(d_k^{(m)})_{k=1}^{\infty}$ defined as

$$a_k^{(m)} = 2^{k-1}m - 1, \tag{5}$$

$$b_k^{(m)} = 2a_{k-1}^{(m)} + b_{k-1}^{(m)} + 1, \tag{6}$$

$$d_k^{(m)} = a_k^{(m)} + b_k^{(m)}. \tag{7}$$

Here

• for $m = 1$, the element $a_0^{(1)}$ is not defined, $a_1^{(1)} = 0$, $b_1^{(1)} = 1$, $a_2^{(1)} = 1$, $b_2^{(1)} = 1$, and the recurrences are applied for $k \geq 3$;

• for odd $m > 1$, we have $a_0^{(m)} = \frac{m-1}{2}$, $b_0^{(m)} = 0$, and the recurrences are applied for $k \geq 2$.

The following formulas hold by a trivial induction:

$$b_k^{(1)} = 2^k - k - 1, \tag{8}$$

$$d_k^{(1)} = 2^k + 2^{k-1} - k - 2, \tag{9}$$

$$b_k^{(m)} = (2^k - 1)m - k + 1, \tag{10}$$

$$d_k^{(m)} = (2^k + 2^{k-1} - 1)m - k. \tag{11}$$

Note that $a_k^{(m)}$ is the number of asymmetric classes $T(i, 2^k m - i) \cup T(2^k m - i, i)$, each of which is nonempty and hence contains at least one orbit of even length. After fringing, each such orbit turns into two orbits lying in the symmetric class $T(2^k m, 2^k m)$. Moreover, the symmetric class $T(2^k m, 2^k m)$ contains orbits obtained by fringing the orbits lying in the previous symmetric class $T(2^{k-1}m, 2^{k-1}m)$, which may have odd length and hence turn into one orbit instead of two. Finally, note that the symmetric class $T(2^k m, 2^k m)$ with $k > 1$ contains trees that are neither fringed nor isocubic, which also form at least one orbit. The recurrence (6) for the sequence $(b_k^{(m)})$ exactly describes all these cases. Thus we have proved the following assertion (for $k = 0, 1, 2$, one can easily prove it directly).

Corollary 6.5. *Let $n = 2^k m$ with m odd. The number of orbits in the class $T(n)$ is at least $d_k^{(m)}$.*

This bound improves the estimate obtained at the end of Sec. 4 for even n: it is slightly smaller than $\frac{3}{2}n$.

7. Empirical data

It is convenient to present empirical data on the number and lengths of orbits of Donaghey's transformation as a generating function. Namely, we consider the polynomial $M_n(x)$ in which the coefficient of x^i equals the number of orbits of Donaghey's transformation of length i on the set of PPCTs with n leaves. The first ten polynomials were given by R. Donaghey in [1]. Strictly speaking, in his formula for $M_8(x)$ the term x^{18} is missing (probably, due to a misprint):

$$\begin{aligned}
M_1(x) &= x,\\
M_2(x) &= x,\\
M_3(x) &= x^2,\\
M_4(x) &= x^2 + x^3,\\
M_5(x) &= x^2 + 2x^6,\\
M_6(x) &= 2x^2 + 3x^3 + x^5 + 4x^6,\\
M_7(x) &= 2x^2 + 14x^6 + x^{20} + x^{24},
\end{aligned} \tag{12}$$

$$M_8(x) = 2x^2 + 7x^3 + 28x^6 + x^9 + x^{10} + x^{15} + x^{18} + 2x^{20} + x^{24} + x^{48} + x^{72},$$
$$\begin{aligned} M_9(x) &= 3x^2 + x^4 + 66x^6 + 2x^{10} + 3x^{12} + 2x^{14} + 5x^{20} \\ &\quad + x^{24} + x^{36} + 2x^{42} + x^{48} + x^{54} + 5x^{72} + x^{90} + x^{144}, \end{aligned}$$
$$\begin{aligned} M_{10}(x) &= 3x^2 + 15x^3 + x^5 + 127x^6 + x^8 + 4x^9 + x^{10} + 10x^{14} + 4x^{18} \\ &\quad + 13x^{20} + x^{24} + x^{27} + 2x^{28} + 2x^{30} + x^{33} + 6x^{36} + 10x^{42} \\ &\quad + x^{45} + 5x^{48} + 4x^{54} + 15x^{72} + 3x^{90} + 2x^{126} + 3x^{144} + x^{147}. \end{aligned}$$

The alternating invariant allows one to consider a more detailed statistic, namely, to take into account not only the size of the trees, but also the value of this invariant. Below we present the corresponding data for trees with 9 and 10 leaves. These tables show that the lower bound on the number of orbits is very rough and should be greatly refined.

Table 1. The orbits of trees with 9 leaves.

Type of the tree	Generating function
(1,8)	$p_1(x) = x^2$
(2,7)	$p_2(x) = x^2 + 9x^6$
(3,6)	$p_3(x) = x^4 + 22x^6 + 2x^{20} + 3x^{72}$
(4,5)	$p_4(x) = x^2 + 35x^6 + 2x^{10} + 3x^{12} + 2x^{14} + 3x^{20} + x^{24} + x^{36} + 2x^{42} + x^{48} + x^{54} + 2x^{72} + x^{90} + x^{144}$

Table 2. The orbits of trees with 10 leaves.

Type of the tree	Generating function
(1,9)	$p_1(x) = x^2$
(2,8)	$p_2(x) = 12x^6$
(3,7)	$p_3(x) = x^2 + 30x^6 + x^{10} + 3x^{20} + x^{24} + x^{36} + 5x^{72}$
(4,6)	$p_4(x) = x^2 + 57x^6 + x^8 + 6x^{14} + 3x^{18} + 8x^{20} + x^{28} + x^{36} + 2x^{42} + 5x^{48} + 3x^{54} + 6x^{72} + 2x^{90} + 2x^{126} + 2x^{144}$
(5,5)	$p_5(x) = 15x^3 + x^5 + 28x^6 + 4x^9 + 4x^{14} + x^{18} + 2x^{20} + x^{27} + x^{28} + 2x^{30} + x^{33} + 4x^{36} + 8x^{42} + x^{45} + x^{54} + 4x^{72} + x^{90} + x^{144} + x^{147}$

Supported by the Federal Target Program "Scientific and scientific-pedagogic personnel of the innovative Russia for 2009–2013," No. 14.B37.21.1641.

Translated by N. V. Tsilevich.

REFERENCES

1. R. Donaghey, "Automorphisms on Catalan trees and bracketing," *J. Combin. Theory, Ser. B.*, **29**, No. 1, 75–90 (1980).
2. D. Callan, "A bijection on Dyck paths and its cycle structure," *Electron. J. Combin.*, **14**, Article R28 (2007).
3. I. A. Pushkarev, "On a transformation of plane trees," *Mat. Vestnik Pedvuzov i Universitetov Volgo-Vyatskogo Regiona*, No. 8, 92–99 (2006).
4. I. A. Pushkarev, "The fractal rotation of fringings of plane cubic trees," *Mat. Vestnik Pedvuzov i Universitetov Volgo-Vyatskogo Regiona*, No. 10, 82–89 (2008).
5. `http://oeis.org/A079438/`.
6. I. Goulden and D. Jackson, *Combinatorial Enumeration*, Wiley, New York (1983).
7. R. Stanley, *Enumerative Combinatorics*, Vol. 2, Cambridge Univ. Press, Cambridge (2005).